SIMATS ENGINEERING
Saveetha Institute of Medical and Technical Sciences
Chennai-602105.

AI-Matched University–Industry Capstone Collaboration Platform with Skill-Gap Recommendation

CSA0925- JAVA PROGRAMMING

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SIMATS Engineering

Project Overview

- The proposed system is a platform connecting university students with industry capstone projects.
- Students create profiles containing their skills, interests, academic background and project preferences.
- Industries can post projects with required technical skills.
- The system matches students with suitable industry projects.
- A skill-gap recommendation module identifies missing skills.
- The platform supports project collaboration and progress tracking.

Main Goal:

To improve student–industry project matching using an intelligent skill-based recommendation approach.

Problem Statement

- Current university–industry collaboration has several challenges:
- Students may not know which industry projects match their skills.
- Companies spend time identifying suitable students.
- Manual project allocation can result in poor skill matching.
- Students may lack some skills required for industry projects.
- There is limited systematic tracking of project progress.
- Skill gaps are often identified only after project allocation.

Problem:

There is a need for an integrated platform that automatically matches students and industry projects while identifying and addressing skill gaps.

Motivation and Need

Motivation

- Improve employability of students.
- Provide students with industry-relevant project experience.
- Reduce manual project allocation.
- Help industries find suitable student teams.
- Encourage skill development.

Need

- The platform provides:
- **Student Profile → Skill Matching → Skill-Gap Detection → Recommendations → Project Collaboration**

EXISTING SYSTEM

- Manual project allocation.
- Faculty recommendations.
- Student self-selection.
- Emails and spreadsheets.
- Separate communication platforms.
- Manual skill comparison.

LIMITATIONS

- Time-consuming.
- Subjective selection.
- Difficult to handle large numbers of students.
- Skill gaps may not be identified.
- No centralized project progress tracking.
- Limited personalized recommendations.

Research / Knowledge Gap

- From the study of existing approaches, the following gaps were identified:
- Lack of integrated student–industry matching.
- Limited automated skill comparison.
- Lack of personalized skill-gap recommendations.
- Poor integration of collaboration and progress tracking.
- Existing systems may focus only on project allocation.

Identified Gap

- A unified platform is required that combines matching, skill-gap analysis, recommendations and project tracking.

Proposed Solution

- The proposed system provides an integrated web-based platform.

Main Functions

- Student registration and profile creation.
- Industry registration.
- Industry project posting.
- Student skill management.
- Automated project–student matching.
- Skill-gap identification.
- Course/resource recommendations.
- Team/project allocation.
- Collaboration and progress tracking.
- Reports and evaluation.

Objectives

Primary Objective

- To develop a Java-based intelligent platform for university–industry capstone collaboration.

Specific Objectives

- Create student and industry profiles.
- Store student skills and project requirements.
- Calculate matching scores.
- Identify missing skills.
- Recommend learning resources.
- Support project allocation.
- Track project progress & Generate project and performance reports.

REQUIREMENTS & SPECIFICATIONS

Functional Requirements

- User registration and login.
- Student profile management.
- Industry profile management.
- Project posting.
- Skill management.
- Project matching.
- Skill-gap analysis.
- Recommendation generation.
- Progress tracking.
- Report generation.

Hardware

- Computer/Laptop
- Minimum 4 GB RAM
- Internet connection
- **Non-Functional Requirements**
- Security
- Reliability
- Scalability
- Usability
- Performance
- Maintainability

Software

- JDK 17/21
- Spring Boot
- MySQL
- Maven
- IDE such as Eclipse/IntelliJ

Literature / Technology Review

Technologies Studied

- Java
- Spring Boot 4.1.0
- Thymeleaf
- MySQL
- Spring data JPA
- Hibernate ORM
- Maven

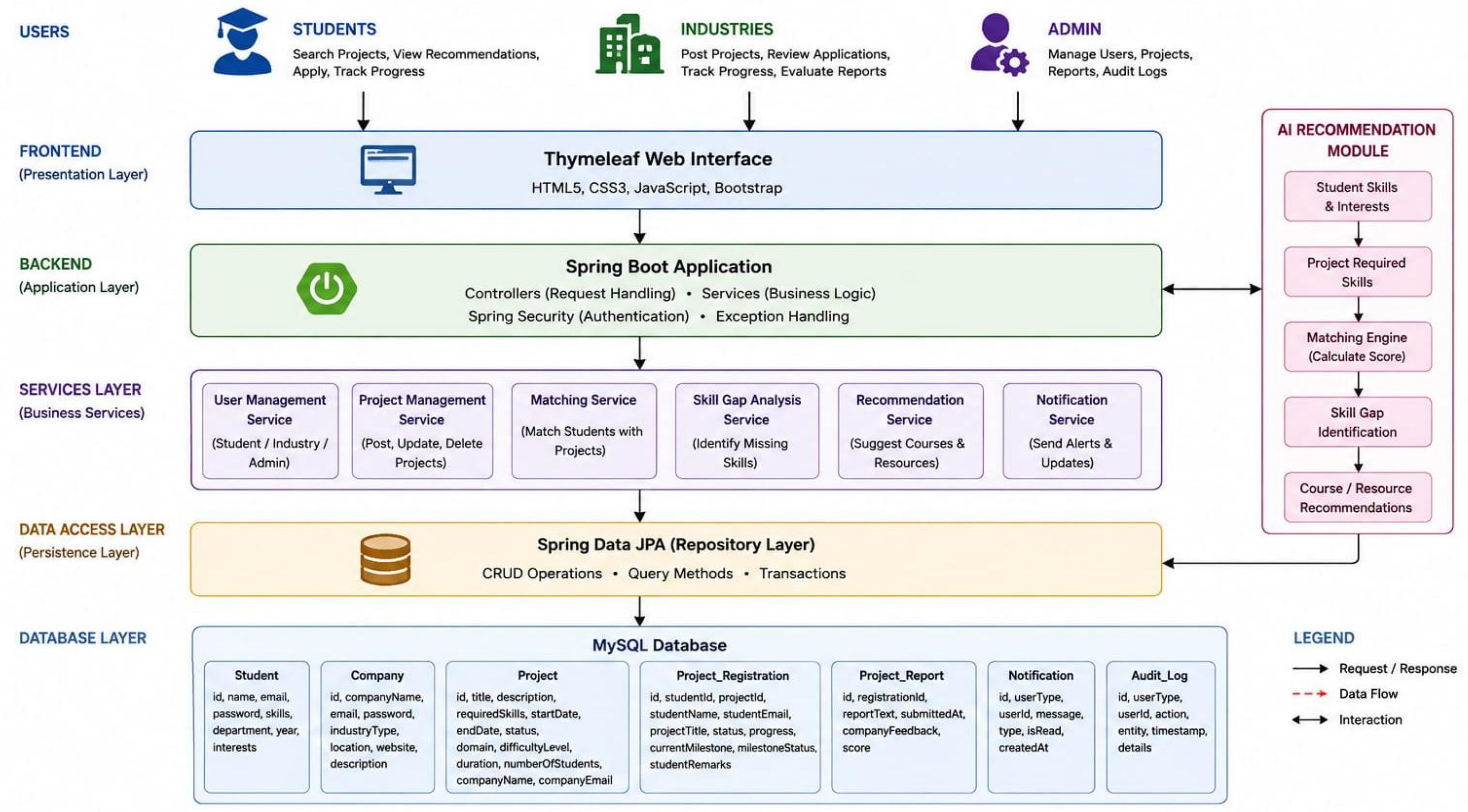
Intelligent Matching

- Skill matching can be performed using:
- Skill similarity
- Weighted matching
- Matching score
- Skill-gap calculation

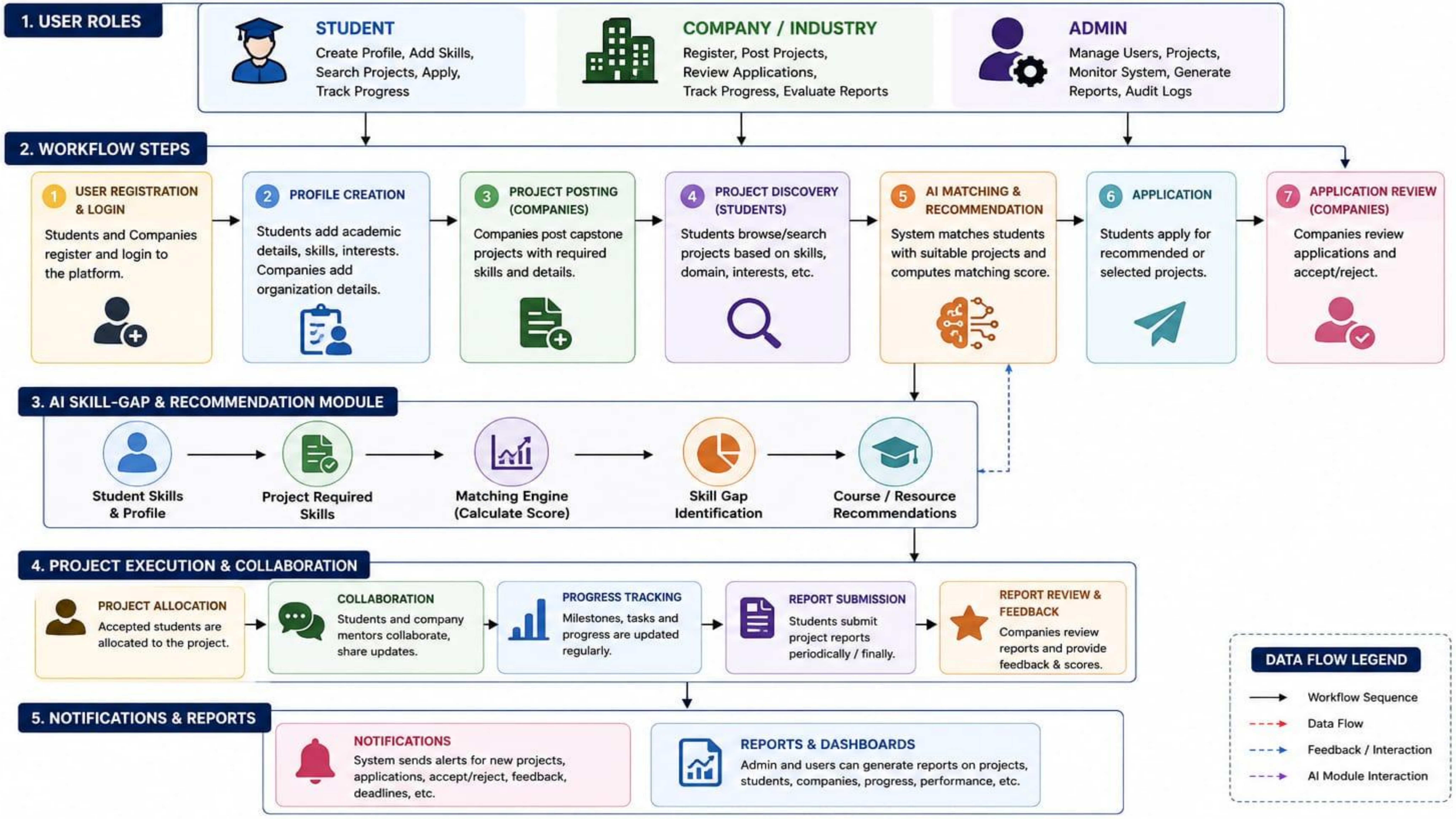
Key Finding

- Combining skill matching with personalized recommendations can improve project allocation and student skill development.

SYSTEM ARCHITECTURE



WORK FLOW


1. USER ROLES
STUDENT
Create Profile, Add Skills, Search Projects, Apply, Track Progress
COMPANY / INDUSTRY
Register, Post Projects, Review Applications, Track Progress, Evaluate Reports
ADMIN
Manage Users, Projects, Monitor System, Generate Reports, Audit Logs
2. WORKFLOW STEPS
1 USER REGISTRATION & LOGIN
Students and Companies register and login to the platform.
2 PROFILE CREATION
Students add academic details, skills, interests. Companies add organization details.
3 PROJECT POSTING (COMPANIES)
Companies post capstone projects with required skills and details.
4 PROJECT DISCOVERY (STUDENTS)
Students browse/search projects based on skills, domain, interests, etc.
5 AI MATCHING & RECOMMENDATION
System matches students with suitable projects and computes matching score.
6 APPLICATION
Students apply for recommended or selected projects.
7 APPLICATION REVIEW (COMPANIES)
Companies review applications and accept/reject.
3. AI SKILL-GAP & RECOMMENDATION MODULE
Student Skills & Profile
Project Required Skills
Matching Engine (Calculate Score)
Skill Gap Identification
Course / Resource Recommendations
4. PROJECT EXECUTION & COLLABORATION
PROJECT ALLOCATION
Accepted students are allocated to the project.
COLLABORATION
Students and company mentors collaborate, share updates.
PROGRESS TRACKING
Milestones, tasks and progress are updated regularly.
REPORT SUBMISSION
Students submit project reports periodically / finally.
REPORT REVIEW & FEEDBACK
Companies review reports and provide feedback & scores.
5. NOTIFICATIONS & REPORTS
NOTIFICATIONS
System sends alerts for new projects, applications, accept/reject, feedback, deadlines, etc.
REPORTS & DASHBOARDS
Admin and users can generate reports on projects, students, companies, progress, performance, etc.
DATA FLOW LEGEND
Workflow Sequence
Data Flow
Feedback / Interaction
AI Module Interaction

Product Design

1. STUDENT

Column Name	Description
id	Unique student ID
name	Student name
email	Student email
password	Login password
department	Department name
year	Academic year
skills	Technical skills
interests	Student interests

2. COMPANY

Column Name	Description
id	Unique company ID
companyName	Company name
email	Company email
password	Login password
industryType	Industry type
location	Company location
website	Company website
description	Company description

3. PROJECT

Column Name	Description
id	Unique project ID
title	Project title
description	Project description
requiredSkills	Required skills
startDate	Project start date
endDate	Project end date
status	Project status
domain	Project domain
difficultyLevel	Difficulty level
duration	Project duration
numberOfStudents	No. of students
companyName	Company name
companyEmail	Company email

4. PROJECT REGISTRATION

Column Name	Description
id	Unique registration ID
studentId	Student ID
projectId	Project ID
studentName	Student name
studentEmail	Student email
projectTitle	Project title
status	Registration status
progress	Overall progress (%)
currentMilestone	Current milestone
milestoneStatus	Milestone status
studentRemarks	Student remarks

5. PROJECT REPORT

Column Name	Description
id	Unique report ID
projectId	Project ID
studentId	Student ID
reportDetails	Report details
submissionDate	Submission date
companyFeedback	Company feedback
score	Score / Marks

6. NOTIFICATION

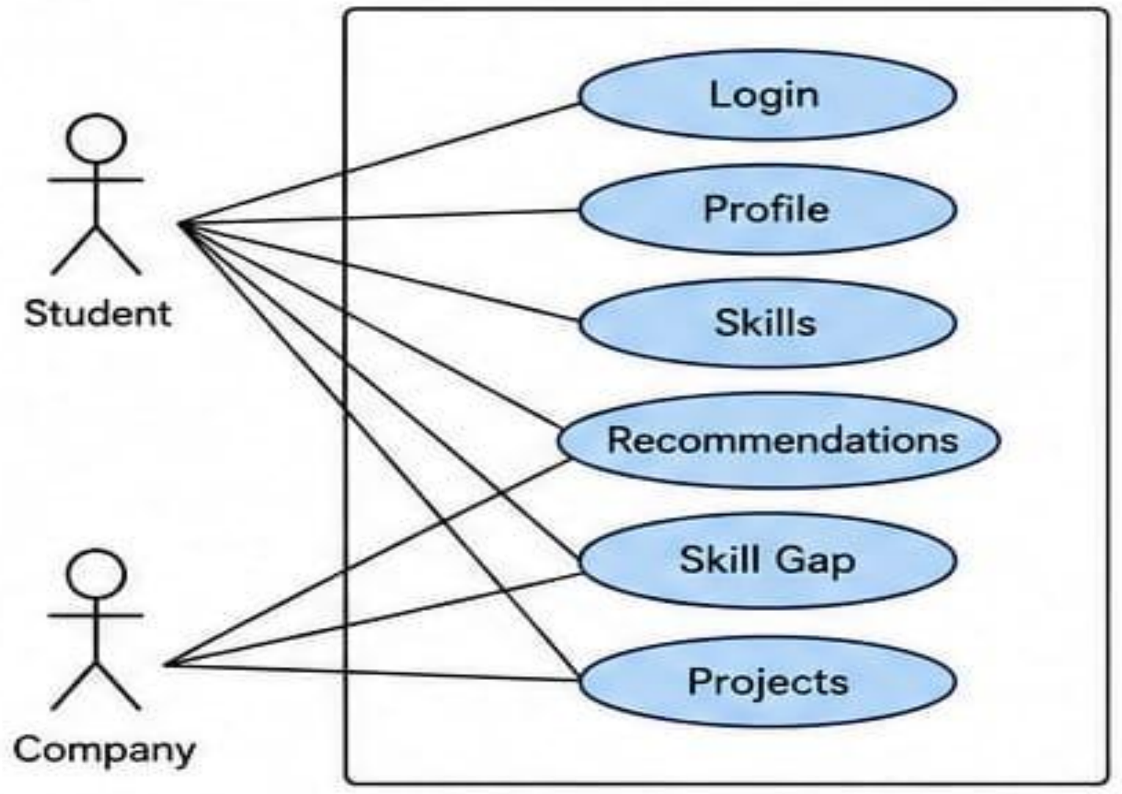
Column Name	Description
id	Unique notification ID
userId	User ID (Student/Company)
userType	User type (Student/Company)
message	Notification message
eventType	Event type
createdAt	Notification date & time
isRead	Read status

7. AUDIT LOG

Column Name	Description
id	Unique log ID
userId	User ID
userType	User type
action	Action performed
tableName	Affected table
recordId	Affected record ID
ipAddress	IP address
createdAt	Date & time

UML & DB

Use Case Diagram

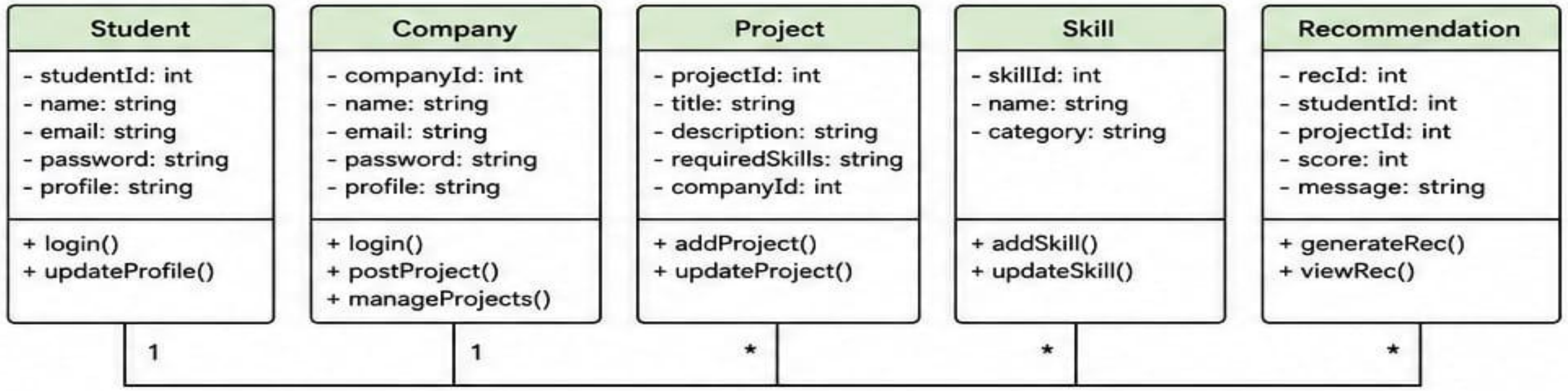


Actors: Student, Company

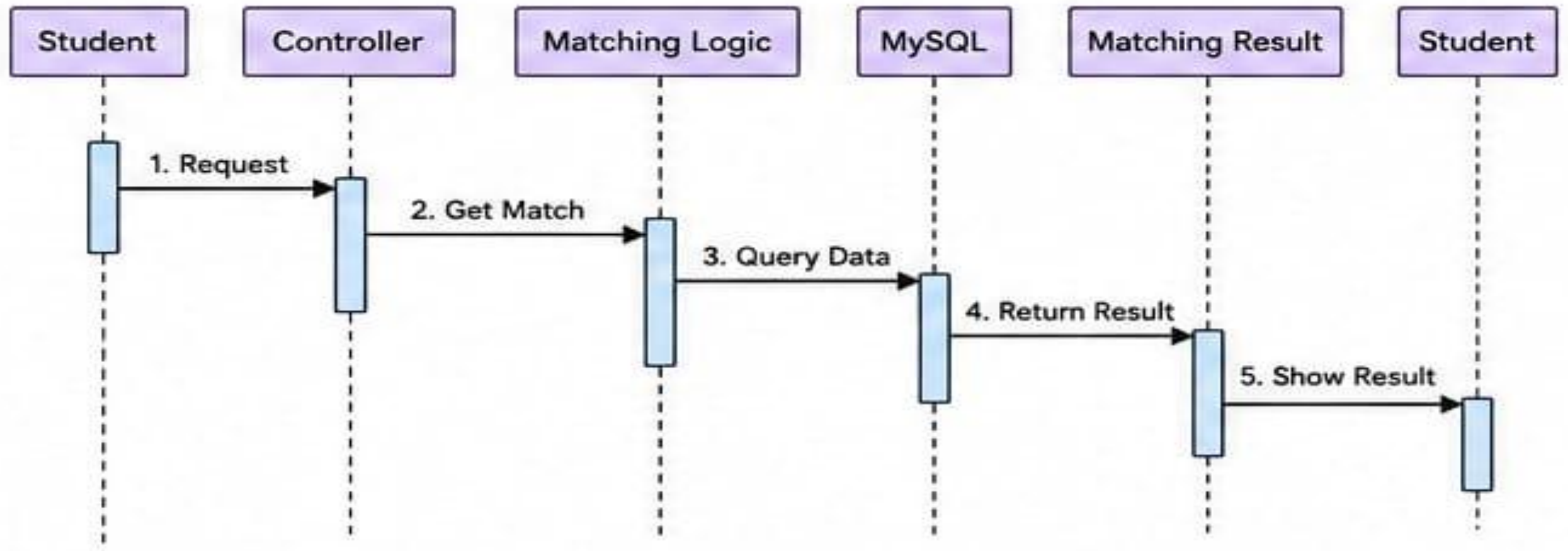
Student: Login, Profile, Skills, Recommendations, Skill Gap, Projects

Company: Login, Profile, Post Projects, Required Skills, Manage Projects

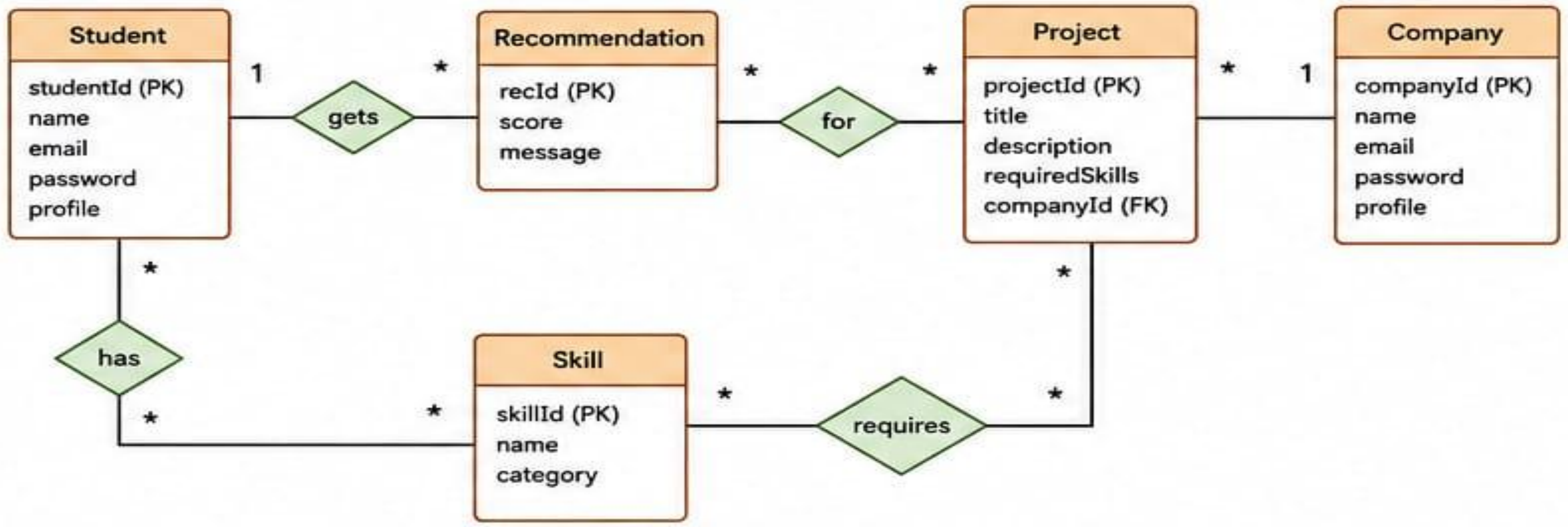
Class Diagram



Sequence Diagram



ER Diagram



TECHNOLOGY USED

Frameworks

- **Spring Boot 4.1.0** — Backend & application development
- **Thymeleaf** — Dynamic web pages
- **Spring Data JPA** — Database interaction

Libraries / Technologies

- **Hibernate ORM** — Object-relational mapping
- **MySQL Connector/J** — Java–MySQL connection
- **Bean Validation** — Input validation

Development Tools

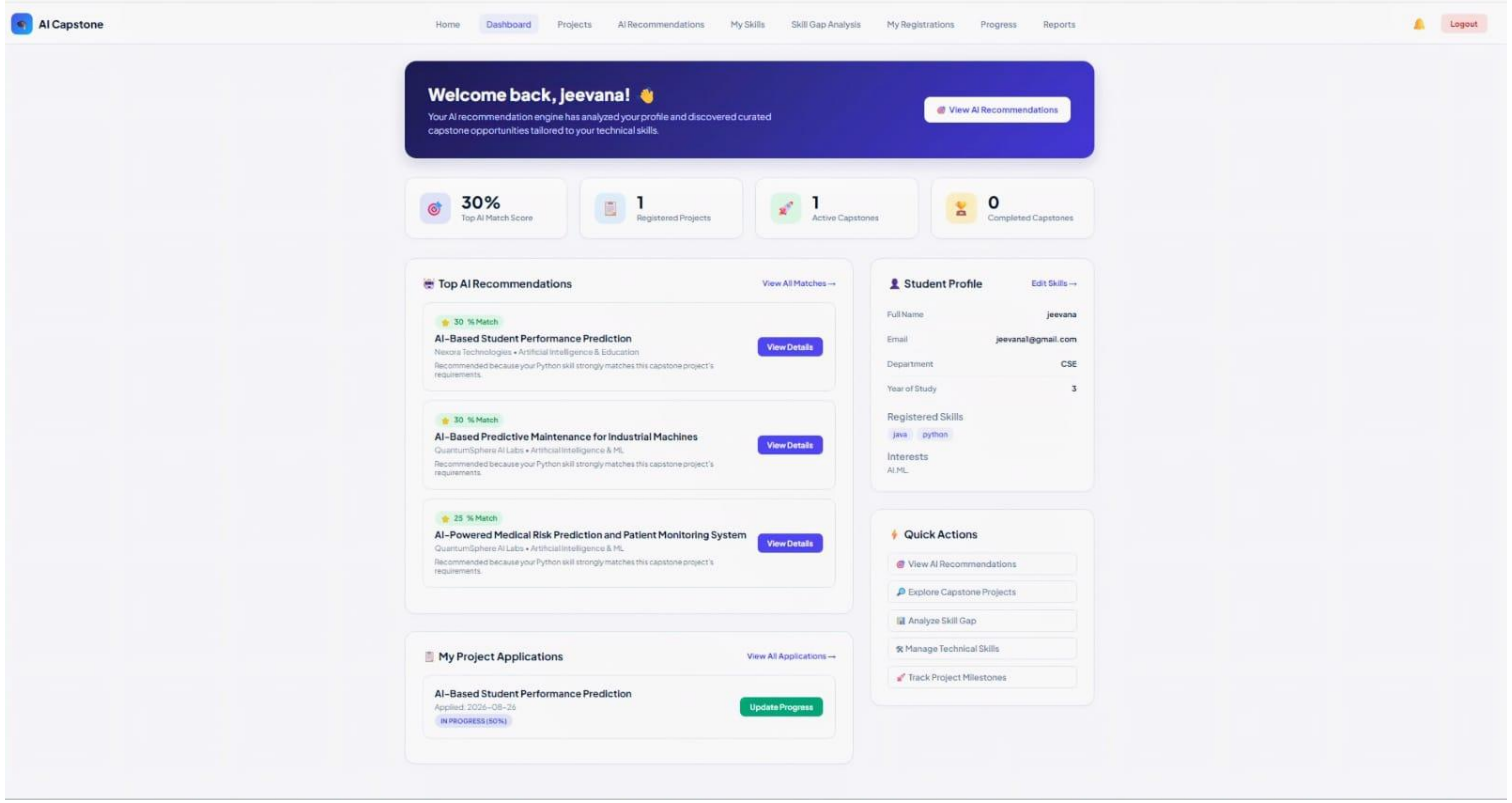
- **IntelliJ IDEA** — Java/Spring Boot development
- **MySQL** — Database management
- **Maven** — Dependency & project management

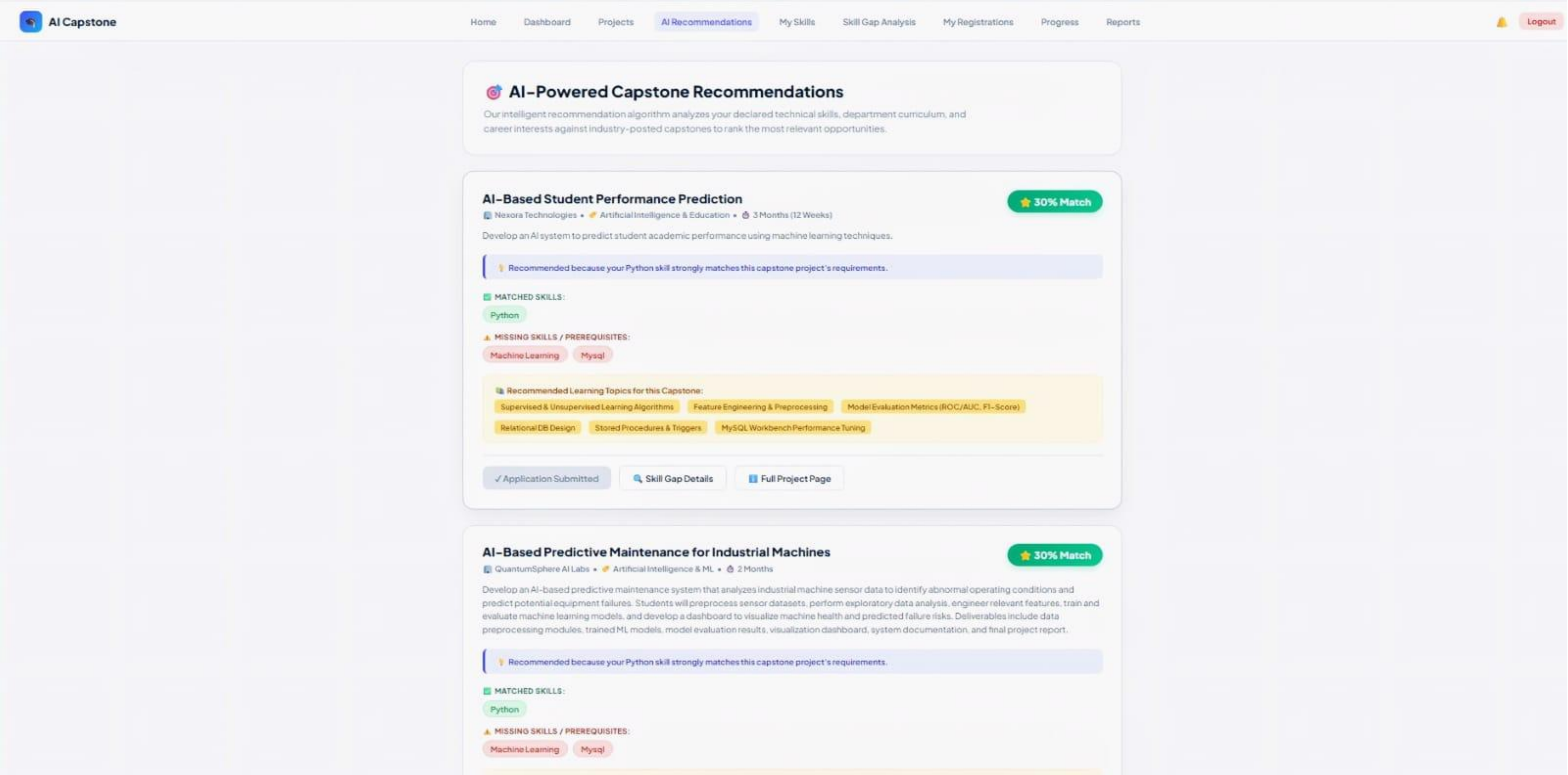
Languages

- **Java 17** — Backend development
- **HTML5** — Web page structure
- **CSS3** — UI styling

Module 1 - Student & AI Recommendation Module

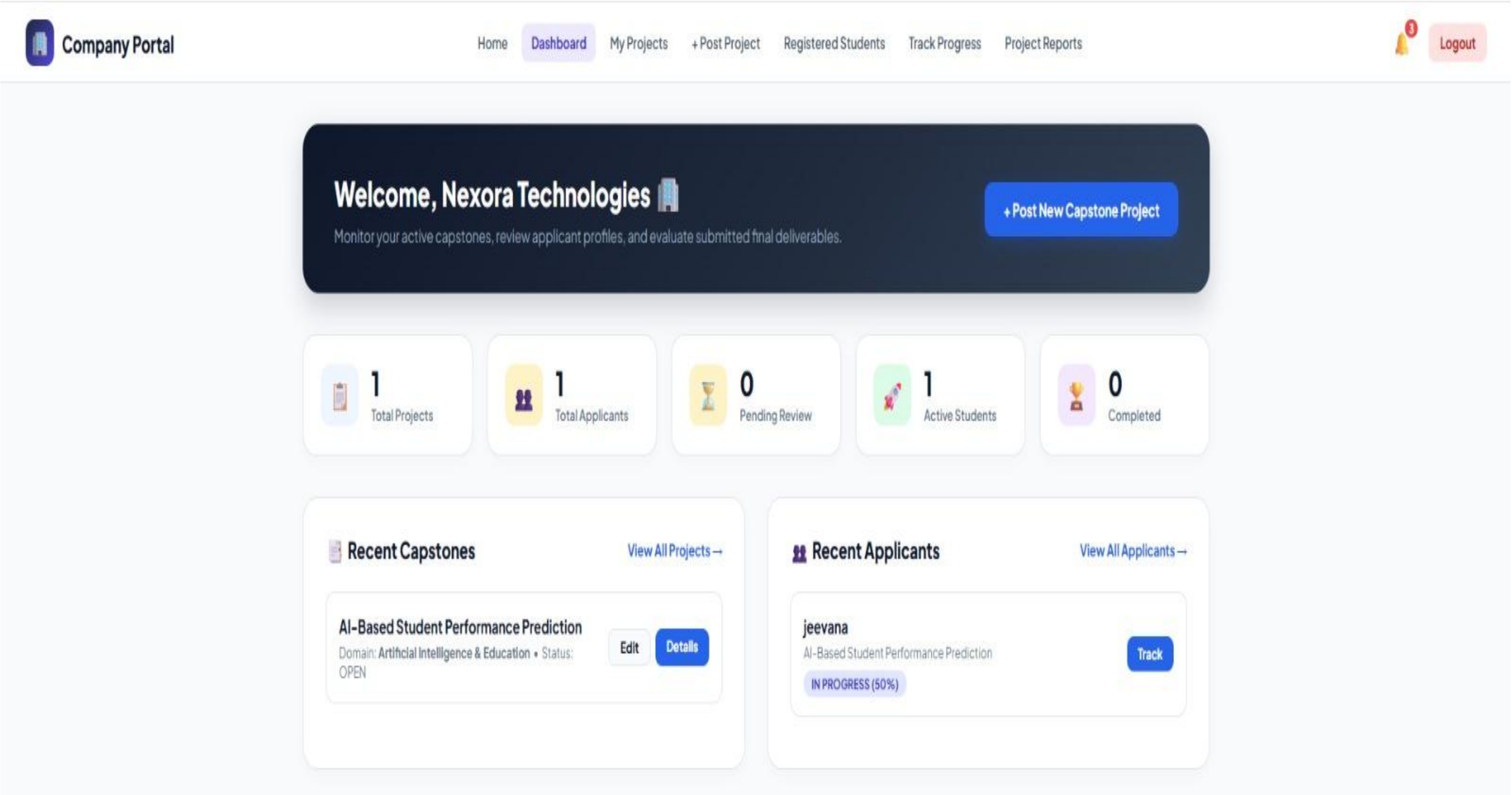
- Student Registration & Login
- Student Profile Management
- Skills & Interests
- AI Project Recommendation
- Skill Gap Analysis
- Personalized Dashboard

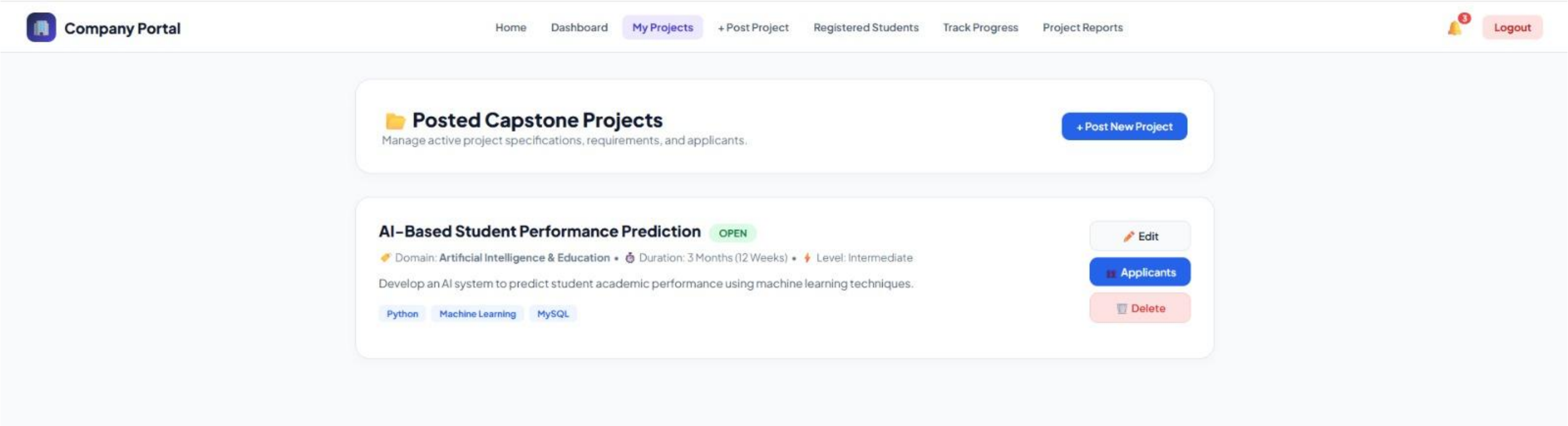


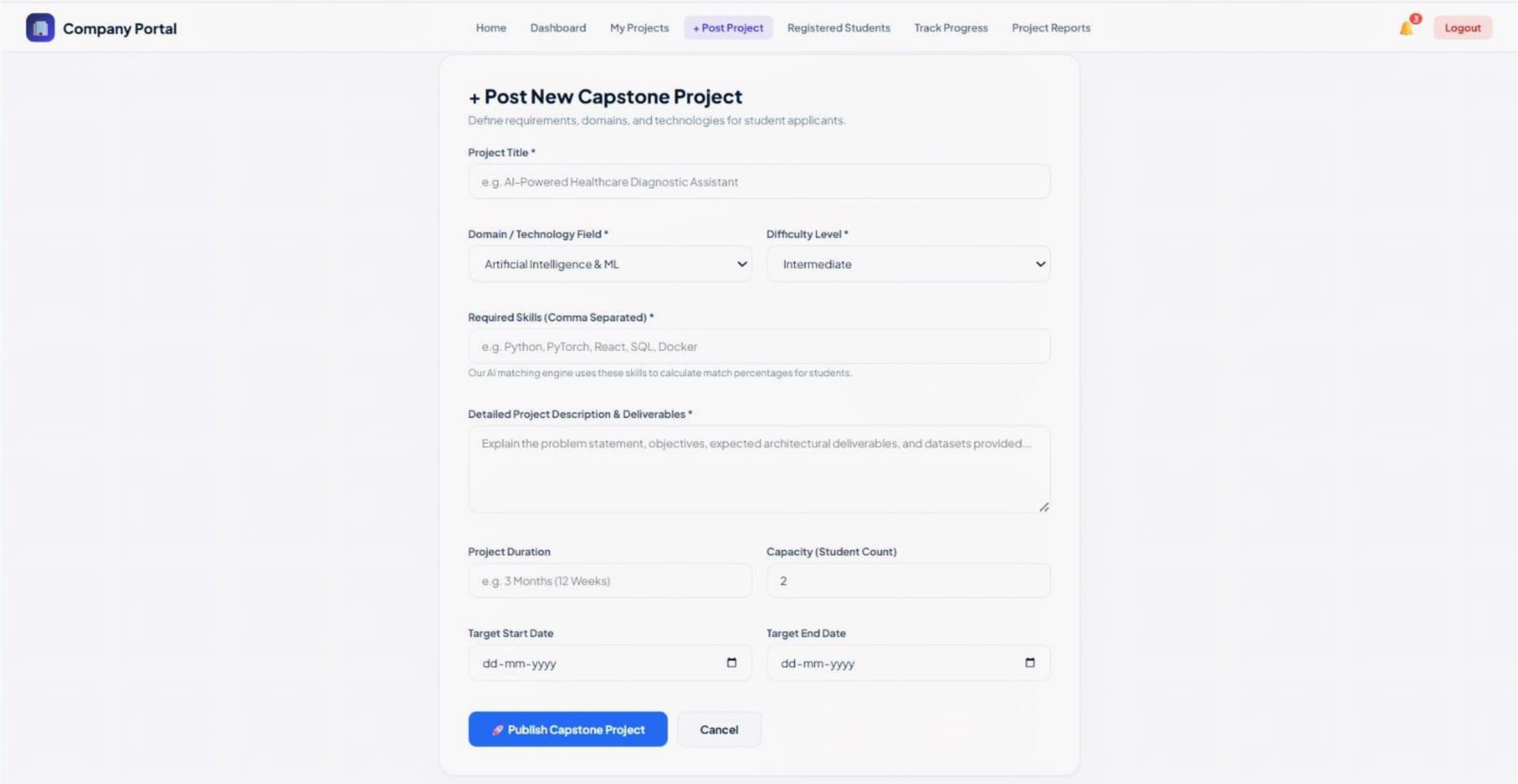

AI Capstone
Home
Dashboard
Projects
AI Recommendations
My Skills
Skill Gap Analysis
My Registrations
Progress
Reports
Logout
AI-Powered Capstone Recommendations
Our intelligent recommendation algorithm analyzes your declared technical skills, department curriculum, and career interests against industry-posted capstones to rank the most relevant opportunities.
AI-Based Student Performance Prediction
30% Match
Nexora Technologies • Artificial Intelligence & Education • 3 Months (12 Weeks)
Develop an AI system to predict student academic performance using machine learning techniques.
Recommended because your Python skill strongly matches this capstone project's requirements.
MATCHED SKILLS:
Python
MISSING SKILLS / PREREQUISITES:
Machine Learning
Mysql
Recommended Learning Topics for this Capstone:
Supervised & Unsupervised Learning Algorithms
Feature Engineering & Preprocessing
Model Evaluation Metrics (ROC/AUC, F1-Score)
Relational DB Design
Stored Procedures & Triggers
MySQL Workbench Performance Tuning
✓ Application Submitted
Skill Gap Details
Full Project Page
AI-Based Predictive Maintenance for Industrial Machines
30% Match
QuantumSphere AI Labs • Artificial Intelligence & ML • 2 Months
Develop an AI-based predictive maintenance system that analyzes industrial machine sensor data to identify abnormal operating conditions and predict potential equipment failures. Students will preprocess sensor datasets, perform exploratory data analysis, engineer relevant features, train and evaluate machine learning models, and develop a dashboard to visualize machine health and predicted failure risks. Deliverables include data preprocessing modules, trained ML models, model evaluation results, visualization dashboard, system documentation, and final project report.
Recommended because your Python skill strongly matches this capstone project's requirements.
MATCHED SKILLS:
Python
MISSING SKILLS / PREREQUISITES:
Machine Learning
Mysql

Module 2 - Company & Project Management

- Company Registration & Login
- Company Profile Management
- Post Capstone Projects
- Define Required Skills
- Manage Project Details
- View Student/Project Information

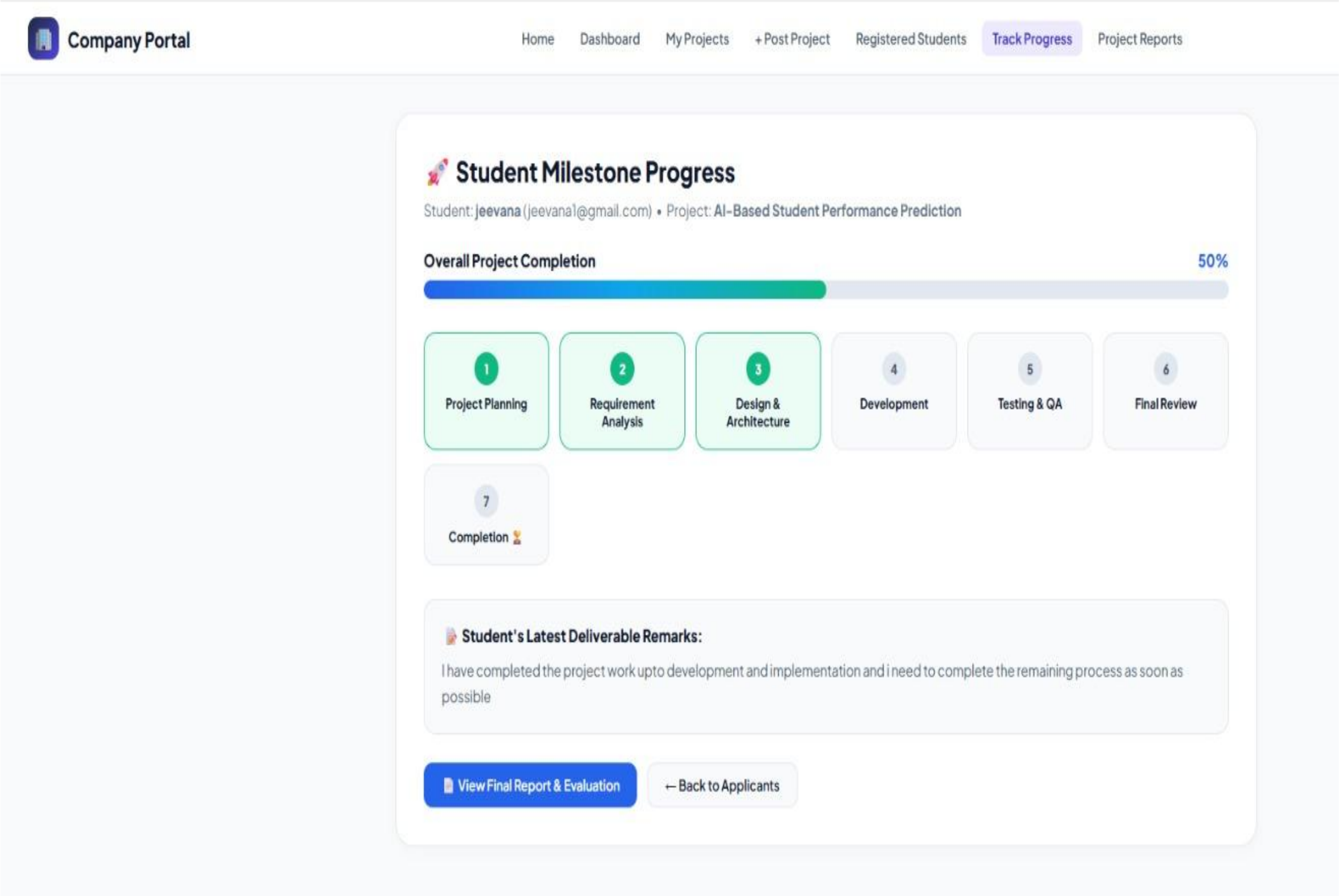



Company Portal
Home
Dashboard
My Projects
+ Post Project
Registered Students
Track Progress
Project Reports
Logout
Posted Capstone Projects
Manage active project specifications, requirements, and applicants.
+ Post New Project
AI-Based Student Performance Prediction
OPEN
Domain: Artificial Intelligence & Education • Duration: 3 Months (12 Weeks) • Level: Intermediate
Develop an AI system to predict student academic performance using machine learning techniques.
Python
Machine Learning
MySQL
Edit
Applicants
Delete

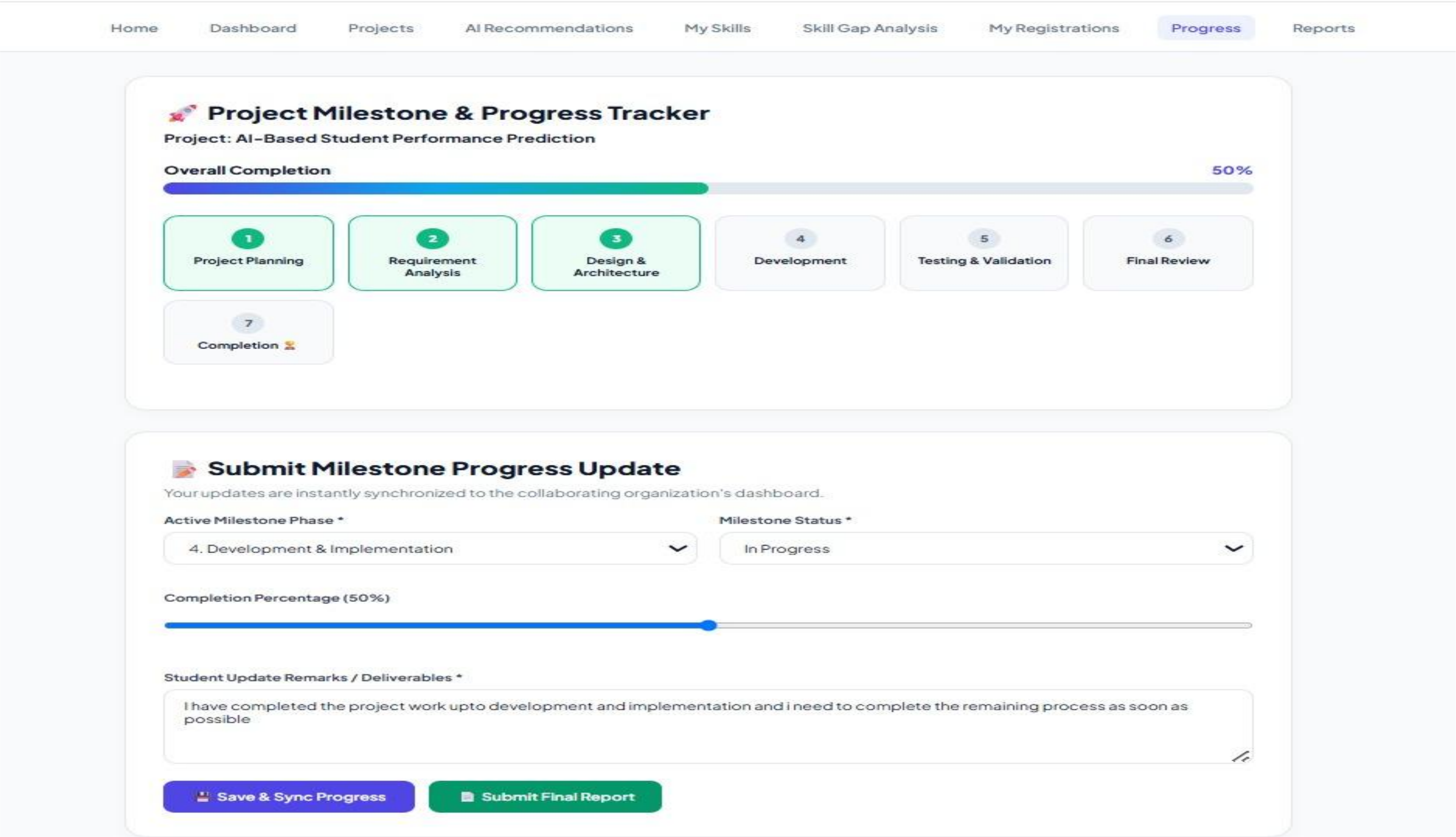

Company Portal
Home
Dashboard
My Projects
+ Post Project
Registered Students
Track Progress
Project Reports
Logout
+ Post New Capstone Project
Define requirements, domains, and technologies for student applicants.
Project Title *
e.g. AI-Powered Healthcare Diagnostic Assistant
Domain / Technology Field *
Artificial Intelligence & ML
Difficulty Level *
Intermediate
Required Skills (Comma Separated) *
e.g. Python, PyTorch, React, SQL, Docker
Our AI matching engine uses these skills to calculate match percentages for students.
Detailed Project Description & Deliverables *
Explain the problem statement, objectives, expected architectural deliverables, and datasets provided...
Project Duration
e.g. 3 Months (12 Weeks)
Capacity (Student Count)
2
Target Start Date
dd-mm-yyyy
Target End Date
dd-mm-yyyy
Publish Capstone Project
Cancel

Module 3- Collaboration & Progress Tracking Module

- Facilitates collaboration between students and company.
- Tracks project milestones, tasks, and deadlines.
- Allows mentors to provide feedback and performance evaluations.
- Monitors student progress through reports and dashboards.
- Supports project report submission and review.



Engineer to Excel



Key Algorithm / Technical Implementation

1. INPUT

Student Skills (with proficiency level)	
Skill	**Proficiency (0-10)**
Java	8
Spring Boot	7
MySQL	6
HTML	7
CSS	5

Student Profile

Project Required Skills (with required level)	
Skill	**Required Leved (0-10)**
Java	7
Spring Boot	8
MySQL	6
REST API	7
JWT	5

Project

2. PROCESS (CONTENT-BASED MATCHING)

Step 1: Fetch student skills and project required skills from MySQL database

Step 2: Find common skills (overlap) between student skills and project required skills

Step 3: Calculate Matching Score

$$\text{Matching Score} = \frac{\sum(\text{Student Level}_i \times \text{Project Level}_i)}{\sum(\text{Project Level}_i)} \times 100$$

Where,

- Student Level_i = Proficiency level of student for skill i
- Project Level_i = Required level of project for skill i

Step 4: Repeat for all projects

Step 5: Sort projects in descending order of Matching Score

3. MATCHING SCORE EXAMPLE

Example:

Student Skills:
{ Java:8, Spring Boot:7, MySQL:6, HTML:7, CSS:5 }

Project Skills:
{ Java:7, Spring Boot:8, MySQL:6, REST API:7, JWT:5 }

Common Skills (Overlap):
{ Java, Spring Boot, MySQL }

Skill	Student Level	Project Level	Product (S x P)
Java	8	7	56
Spring Boot	7	8	56
MySQL	6	6	36

Sum of Products = 148

Sum of Project Levels = 7 + 8 + 6 + 7 + 5 = 33

Matching Score = (148 / 33) × 100 = 448.48% → 100% (capped)

Final Matching Score = 100%

4. SKILL-GAP ANALYSIS (ADDITIONAL FEATURE)

Purpose: Identify missing or low-level skills required for a selected project.

For each required skill:

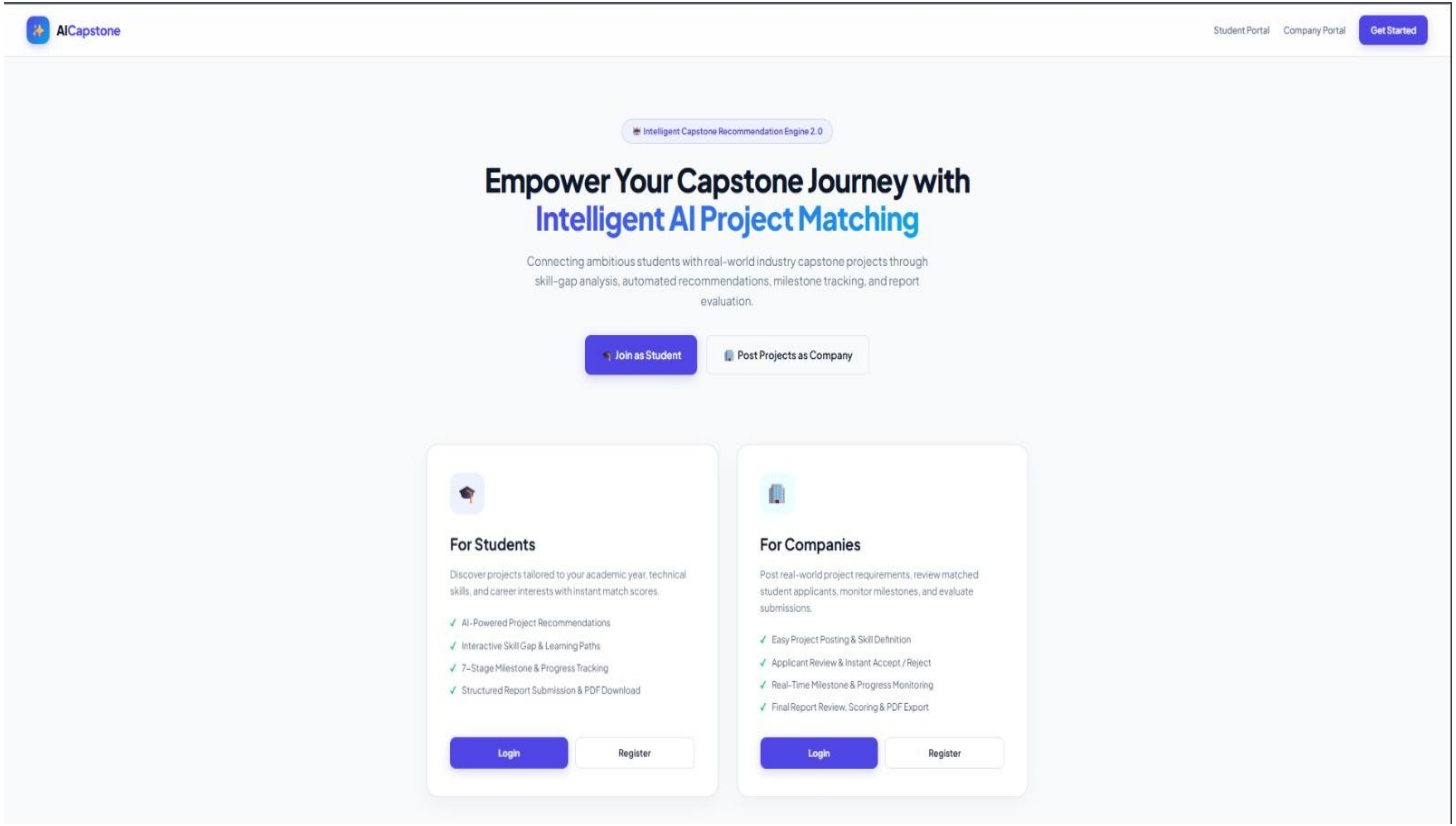
- If skill not present in student skills → **Missing Skill**
- If student level < required level → **Skill Gap**
- Else → **No Gap**

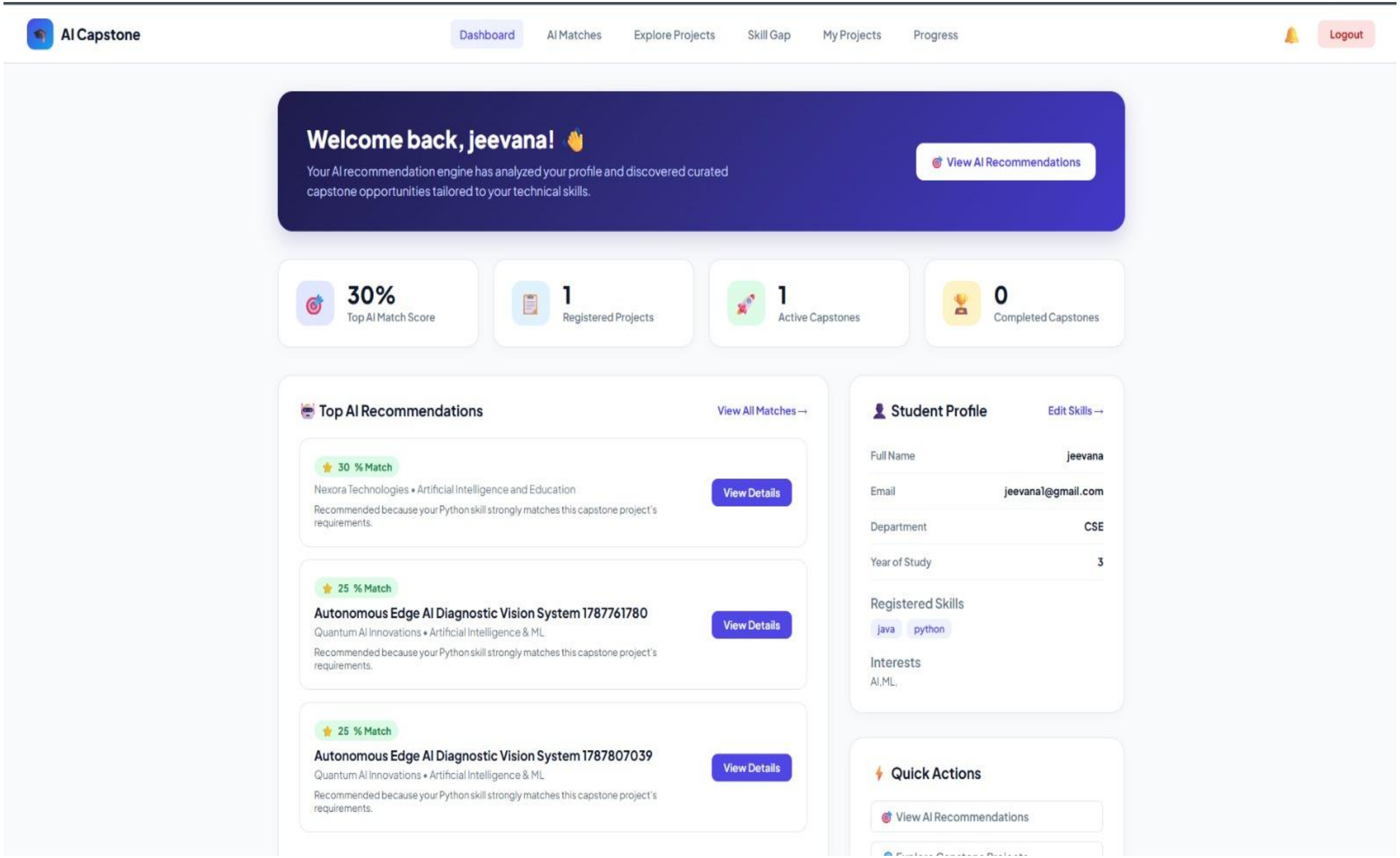
Example (for the above project):

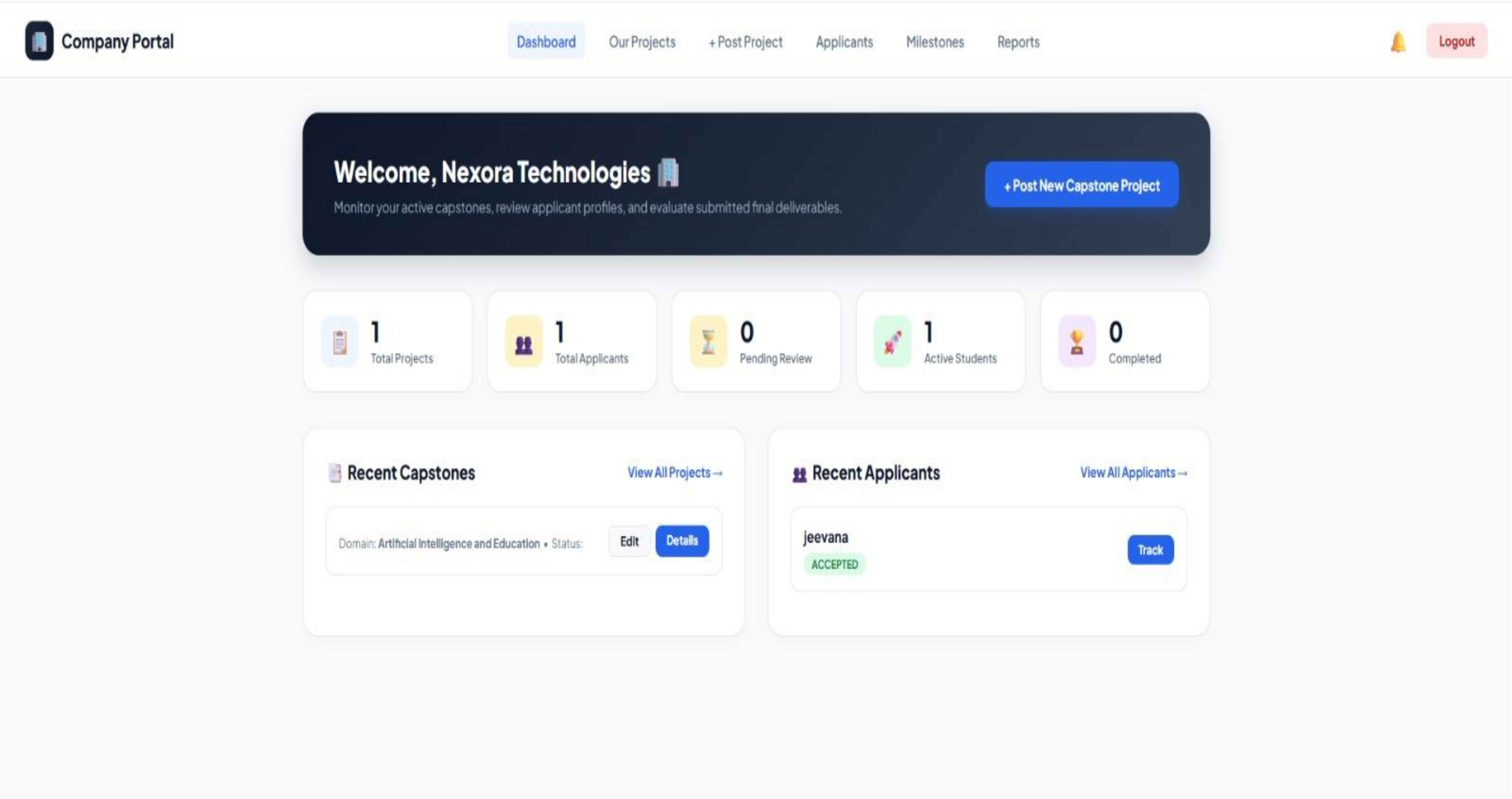
Skill	Required Level	Student Level	Gap	Status
Java	7	8	0	No Gap
Spring Boot	8	7	1	Skill Gap
MySQL	6	6	0	No Gap
REST API	7	-	7	Missing Skill
JWT	5	-	5	Missing Skill

Skill Gap Output:
Improve in: Spring Boot (gap: 1), Learn: REST API, JWT

PROTOTYPE/ WORKING MODULE




AI Capstone
Dashboard
AI Matches
Explore Projects
Skill Gap
My Projects
Progress
Logout
Welcome back, jeevana!
Your AI recommendation engine has analyzed your profile and discovered curated capstone opportunities tailored to your technical skills.
View AI Recommendations
30%
Top AI Match Score
1
Registered Projects
1
Active Capstones
0
Completed Capstones
Top AI Recommendations
View All Matches →
30 % Match
Nexora Technologies • Artificial Intelligence and Education
Recommended because your Python skill strongly matches this capstone project's requirements.
View Details
25 % Match
Autonomous Edge AI Diagnostic Vision System 1787761780
Quantum AI Innovations • Artificial Intelligence & ML
Recommended because your Python skill strongly matches this capstone project's requirements.
View Details
25 % Match
Autonomous Edge AI Diagnostic Vision System 1787807039
Quantum AI Innovations • Artificial Intelligence & ML
Recommended because your Python skill strongly matches this capstone project's requirements.
View Details
Student Profile
Edit Skills →
Full Name
jeevana
Email
jeevana1@gmail.com
Department
CSE
Year of Study
3
Registered Skills
java
python
Interests
AI,ML,
Quick Actions
View AI Recommendations


Company Portal
Dashboard
Our Projects
+ Post Project
Applicants
Milestones
Reports
Logout
Welcome, Nexora Technologies
Monitor your active capstones, review applicant profiles, and evaluate submitted final deliverables.
+ Post New Capstone Project
1
Total Projects
1
Total Applicants
0
Pending Review
1
Active Students
0
Completed
Recent Capstones
View All Projects →
Domain: Artificial Intelligence and Education • Status:
Edit
Details
Recent Applicants
View All Applicants →
jeevana
ACCEPTED
Track

TESTING & VALIDATION

Test Case ID	Test Scenario & Objective	Input / Conditions	Expected System Output	Actual Observed Result	Status
TC-01	**Student Registration** Verify that a new student can register successfully.	Valid details: Name, Email, Password, Department, Skills	Student account created successfully and verification email sent.	Student registered and redirected to login page correctly.	PASSED
TC-02	**User Login & Role Access** Verify login and role-based access for all users.	Valid credentials for Student / Company / Admin	User logged in and redirected to respective dashboard with proper permissions.	Login successful. Correct dashboard and menus displayed.	PASSED
TC-03	**Company Project Posting** Verify that a company can post a new project.	Logged-in company; project title, description, required skills	Project posted successfully and visible to students.	Project created and listed in project directory.	PASSED
TC-04	**Skill Management** Verify that student can add and update skills.	Student logged in; add skills (e.g., Java, MySQL, Spring Boot)	Skills saved and displayed correctly in student profile.	Skills added and reflected in profile correctly.	PASSED
TC-05	**AI Project Recommendation** Verify AI recommends suitable projects based on skills.	Student skills available; AI matching algorithm executed	Top relevant projects recommended with match percentage.	Top 5 projects shown with match scores.	PASSED
TC-06	**Skill-Gap Analysis** Verify that missing skills are identified.	Student views a project detail with required skills	Missing skills and learning resources suggested correctly.	Skill gaps and resources displayed correctly.	PASSED
TC-07	**Project Application** Verify that student can apply for a project.	Student applies for selected project	Application submitted successfully and status shown as 'Pending'.	Application submitted and visible to company.	PASSED
TC-08	**Application Decision (Company)** Verify company can accept or reject application.	Company views applications and takes action	Status updated to 'Accepted' or 'Rejected' and student notified.	Application accepted and notification sent to student.	PASSED
TC-09	**Progress Tracking** Verify project progress and milestones can be updated.	Project started; update milestone/progress	Progress percentage and milestone status updated correctly.	Progress and milestone updated correctly.	PASSED
TC-10	**Report Submission & Review** Verify student can submit report and company can review.	Student uploads final report; company reviews and gives feedback	Report submitted and feedback saved successfully.	Report uploaded, reviewed and feedback stored in database.	PASSED
TC-11	**Notification System** Verify real-time notifications are sent to users.	Trigger events: application, acceptance, feedback, etc.	Notification generated and displayed to the user.	Notification received and shown in dashboard.	PASSED

RESULTS & PERFORMANCE

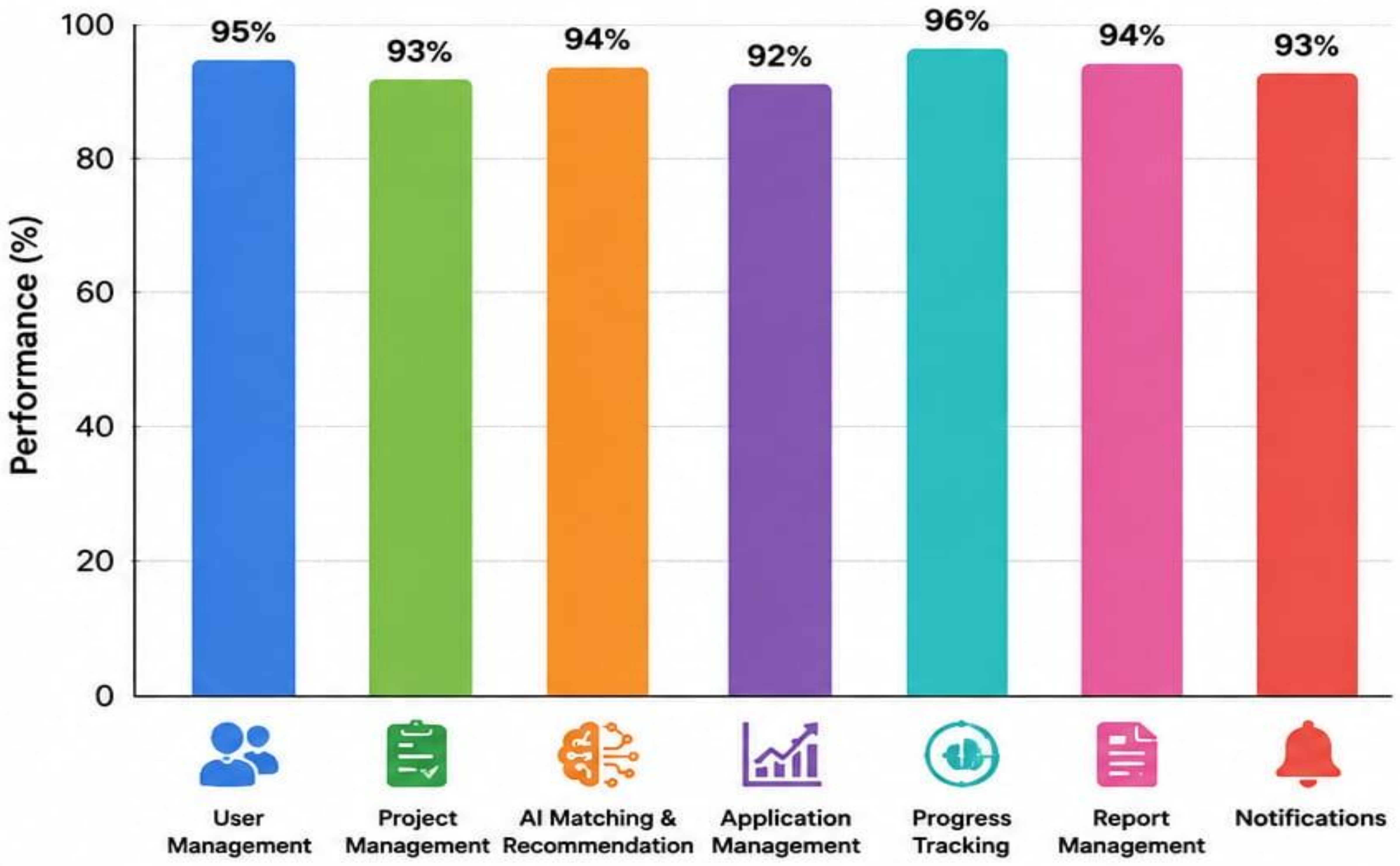
RESULTS

- Developed a complete Capstone Project Management Platform with Student, Company and Admin modules.
- **AI based project recommendation** matches students with suitable projects based on skills.
- **Skill-gap analysis** identifies missing skills and recommends resources for improvement.
- Companies can post projects, manage applications, **track progress** and **review reports**.
- Students can apply for projects, **track progress**, submit reports and **receive notifications**.
- **Role-based access control** ensures secure and efficient management of all operations.

PERFORMANCE SUMMARY

- All major modules are functioning as expected.
- System tested for core workflows and role-based operations.
- Fast response time for project recommendations and application processing.
- Reliable data management with MySQL database.
- User interface is responsive and easy to use.

MODULE PERFORMANCE OVERVIEW



Overall Outcome: The system successfully achieves efficient management of capstone projects and enhances industry-student collaboration.

COMPARISON BETWEEN EXISTING SYSTEM AND PROPOSED SYSTEM

FEATURE	EXISTING SYSTEM	PROPOSED SYSTEM
Project Matching	Manual matching	AI-based automatic matching
Skill Gap Analysis	Not available	Automatic skill gap analysis
Recommendations	Limited or generic	AI-based personalized recommendations
Dashboard	Basic dashboard	Personalized dashboard
User Access	Single access	Separate access for Student & Company
Progress Tracking	Manual tracking	Automatic tracking with milestones
Data Management	Scattered / Spreadsheets	Centralized MySQL database
Security	Basic security	Role-based secure access
Efficiency	Low efficiency	High efficiency

INDIVIDUAL CONTRIBUTIONS – 2 STUDENTS

STUDENT	MODULE / WORK	CONTRIBUTION
D. Jeevana Reddy 192411259	01 Module 1: Student & AI Recommendation	Student registration & login, student profile, skills & interests Personalized dashboard AI project recommendation Skill-gap analysis
M. Chandana 192472250	02 Module 2: Company & Project Management	Company registration & login, company profile Project posting Required skills Project management Student/project matching information
Both Students	03 Module 3: Collaboration & Progress Tracking	Project tasks, milestones, deadlines Mentor feedback Progress tracking Notifications and reports

Challenges & Engineering Decisions

Challenges

- Designing a suitable matching mechanism.
- Managing student and project skills.
- Avoiding duplicate data.
- Designing secure authentication.
- Connecting frontend and backend.
- Maintaining project progress information.

Engineering Decisions

- Java selected for backend implementation.
- Spring Boot selected for for backend web application development and request handling.
- MySQL selected for structured relational data.
- Hibernate/JPA used for database interaction.
- JWT used for secure authentication.

Limitations

- Current limitations may include:
- Matching quality depends on the accuracy of student skill
- Limited number of learning resources.
- Recommendation quality can be improved with more data.
- Initial version may support only selected project domains.
- Real-world deployment requires larger datasets.
- Advanced AI models can improve future matching accuracy.

Future Enhancements

- Advanced machine-learning-based matching.
- Resume analysis.
- Automatic skill extraction.
- More personalized learning recommendations.
- Mobile application.
- Industry feedback analytics.
- Advanced dashboards.

Conclusion

- Connects students and companies for capstone projects.
- Provides project recommendations and skill-gap analysis.
- Supports project applications and acceptance/rejection.
- Enables progress tracking and report management.
- Provides role-based access for students, companies, and admins.
- Improves industry–student collaboration.

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THANK YOU